

Why a satisfied dual still violates the limit

WITHOUT A PROBE	WITH PROBE AND MARGIN
the dual is fed $\hat{J}_C$ measured in G_θ	probe rolls π_ϕ in F every N iterations
violation is realised under the true F	plan against d_{eff} , not against d
nothing in the loop ever compares them	13 of 15 seed runs entirely clean
7.3 percent of evaluations violate	about 1 percent, and a stated rate

The margin, and the only property it needs

$$e_i = c_i - \bar{c} \quad \text{on the } n \text{ real episode costs the probe returned}$$
$$q = e_{(\lceil (n+1)(1-\delta) \rceil)} \qquad d_{\text{eff}} = d - (q + \hat{b})$$

exchangeability of the deviations gives $\mathbb{P}[c > d] \leq \delta$

A margin is only as good as the distribution it bounds

The first attempt conformalised the surrogate's cost error. On the reaction diffusion family the surrogate is accurate and the policy is variable, so that quantile never saw what violates the limit: violations rose to 18.2 percent against a 14.5 percent baseline. Conformalising the episode costs holds the bound.

